

SF1811 Optimization

Lecture 1: Introduction to optimization and linear programming

Chapters 1–3

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Introduction to optimization and linear programming

Main question:

How can a practical decision problem be formulated
as a linear programming problem?

Today:

- Optimization problems.
- The Tables and chairs problem.
- Geometric interpretation.
- Linear programming in standard form.

Optimization problem

Basic form:

$$\begin{aligned} &\text{minimize} && f(x) \\ &\text{subject to} && x \in F. \end{aligned}$$

- x: decision variables,
- f: objective function,
- F: feasible region.

A point $x \in F$ is a feasible solution.

An optimal solution is a feasible solution with the best objective value.

$$\underset{x \in F}{\text{maximize}} \; f(x) \quad \Longleftrightarrow \quad \underset{x \in F}{\text{minimize}} \; -f(x).$$

Tables and chairs

A manufacturer produces tables and chairs.

	Large parts	Small parts	Revenue
One table	1	2	SEK 400
One chair	1	1	SEK 300

Available resources:

- 200 large parts,
- 300 small parts.

How many tables and chairs should be produced to maximize total revenue?

Formulation of Tables and chairs

Introduce the decision variables

$$T = \text{number of tables}, \qquad C = \text{number of chairs}.$$

- Objective function:
- Resource constraints:
- Sign restrictions:

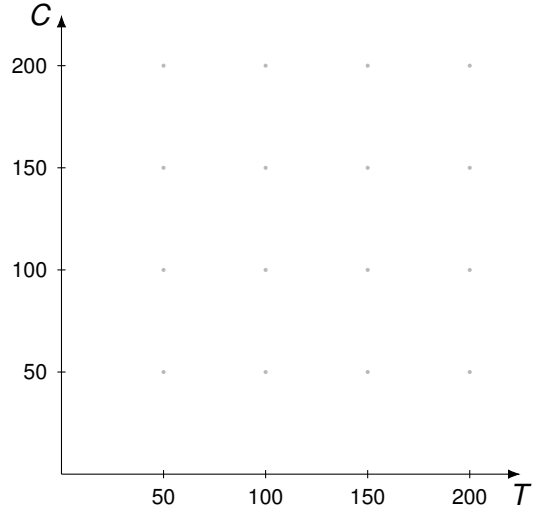
Geometric interpretation

Constraints:

$$T + C \leq 200,$$
$$2T + C \leq 300,$$
$$T, C \geq 0.$$

Objective function:

$$400T + 300C.$$



Linear programming in standard form

Component form:

$$\begin{aligned} \text{minimize} \quad & \sum_{j=1}^{n} c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^{n} a_{ij} x_j = b_i, \quad i = 1, \ldots, m, \\ & x_j \geq 0, \qquad\qquad j = 1, \ldots, n. \end{aligned}$$

Vector form:

$$\begin{aligned} \text{minimize} \quad & c^{\mathsf{T}} x \\ \text{subject to} \quad & Ax = b, \\ & x \geq 0. \end{aligned}$$

This form is referred to as standard form.

Slack variables

An inequality

$$a_i^T x \leq b_i$$

can be written as

$$a_i^T x + s_i = b_i, \qquad s_i \geq 0.$$

Hence,

$$s_i = b_i - a_i^T x.$$

The slack variable s_i measures the remaining margin in constraint i.

Tables and chairs in standard form

Introduce slack variables s_1 and s_2.

The maximization problem is written as a minimization problem by changing the sign of the objective function.

From geometry to algebra

At different vertices, different original variables or slack variables are equal to zero.

Vertex (T, C)	Variables equal to zero
$(0, 0)$	
$(0, 200)$	
$(100, 100)$	
$(150, 0)$	

Can the vertices be generated algebraically by choosing which variables to set equal to zero?

Main ideas

After this lecture, you should be able to:

- identify decision variables, objective function and constraints;
- formulate a simple linear programming model;
- interpret a two-variable linear program geometrically;
- introduce slack variables;
- write a linear program in standard form;
- interpret a slack variable as remaining margin.

Next lecture:

Why can an optimal solution be sought among basic feasible solutions?